

Introduction to Probability and Statistics

Exam June 2024

Read carefully the exam problems, noting that those with a higher degree of difficulty have been marked with asterisks “***”.

Problem P.1

A web-site access is protected by a password that consists of 5 digits that could be either 0 or 1. A hacker does not know the password and tries to guess this random combination.

- 1) What is the probability that the hacker guesses right from the first attempt?
- 2) What is the probability that the hacker guesses right on the third attempt?
- 3) What is the probability that it will take 3 or less attempts to guess the password?

Problem P.2

The cumulative distribution function for a random variable (RV) X is given by the following formula:

$$F(x) = \begin{cases} 0, & x \leq 2 \\ \frac{x}{2} - 1, & 2 < x \leq 4 \\ 1, & x > 4 \end{cases}$$

- 1) Calculate the following probabilities:
 - a) $P(X < 2.5)$
 - b) $P(X \leq 2.5)$
 - c) $P(X > 2.5)$
 - d) $P(X = 5)$
- 2) Find the probability density function (PDF) for RV X . Does this distribution have a special name?
- 3) Find $E[X]$, $\text{Var}(X)$, $E[X^2 + \exp(X)]$.

Problem P.3

An RV X is normally distributed with mean 20. It is given that the probability that X lies within the interval from 20 to 40 is equal to 0.3 but the standard deviation is unknown.

- 1) Find the probability that X lies in the interval from 0 to 20.
- 2) Find the probability $P(X > 40)$.
- 3) *** Find the standard deviation of X .
- 4) *** Find the probability that X lies in the interval from 5 to 20.

Problem S.1

In Denmark, approximately 0.4 of the population cycle to work and school. If you were to estimate the exact fraction of the danish population that cycle to work and school using a questionnaire (yes or no answer):

- 1) Where should you distribute this questionnaire?
 - a) Aalborg
 - b) Aarhus
 - c) Copenhagen
 - d) All of the above

Justify your answer

- 2) How many questionnaire answers do you need to collect to guarantee that the 90% confidence interval of your estimate $\hat{p}$ is inside the range $(\hat{p} - 0.01, \hat{p} + 0.01)$?

Problem S.2

You want to characterise the parameters that determine the power consumption of a wireless transmitter. You were told that there is a linear relation between the transmission power P_{tx} and the total power consumed by the circuit P_{circuit} , given as

$$P_{\text{circuit}} = \beta_1 P_{\text{tx}} + \beta_0,$$

where β_0 is the fixed power needed to turn on the circuit and β_1 is the amplifier's inefficiency. To estimate those parameters, you collected a series of measurements that are affected by Gaussian noise, so your linear model is

$$P_{\text{measured}} = \beta_1 P_{\text{tx}} + \beta_0 + P_{\text{noise}},$$

where P_{noise} is the power of the noise, which is a Gaussian random variable with parameters $(0, \sigma^2)$.

You collected a huge amount of measurements for different transmission powers and (supposedly) saved them in your PC but you cannot find them anywhere. So, the only option left is to estimate the parameters using a few measurements that you recorded in your notebook, shown in Table I.

TABLE I
MEASURED POWER CONSUMPTION OF THE CIRCUIT FOR DIFFERENT SETTINGS OF THE TRANSMISSION POWER.

Transmission power P_{tx} [W]	0.1	0.2	0.5	0.7	0.9
Measured power P_{measured} [W]	0.43	0.52	1.13	1.73	2.01

- 1) Estimate the parameters β_0 and β_1
- 2) Predict the average measured power for $P_{\text{tx}} = 1$ W
- 3) *** Can you reject the null hypothesis $H_0: \beta_1 \leq 1.5$ with 5% significance? Report the test statistic or p-value to support this conclusion.

Problem S.3

Consider the following hypothetical scenario. A website has compiled a list with the battery lifetime for a large collection of mobile phone models. To make this list, the website runs a battery lifetime test app and records the time it takes for the phone to go from 100% charge until it shuts down. This process is repeated many times and the average battery lifetime is reported for each phone, which can be taken as the ground truth. The longest battery lifetime for any phone in the market is 1120 minutes.

A new phone model is released and the company behind it claims that it is the one with longest battery lifetime in the market. The website is set to release a review article about this new phone and needs to test whether the claim of the company is true but, due to time constraints, it can only perform the test on the phone 5 times and gets the results shown in Table II.

TABLE II
BATTERY LIFETIME OF THE NEW PHONE IN THE TESTS.

Test number	1	2	3	4	5
Battery lifetime (minutes)	1090	1130	1122	1101	1110

- 1) Assume that the website knows that the battery lifetime of the phones tested with the app follows a normal distribution with parameters $(\mu_{\text{phone}}, \sigma^2)$ with $\sigma = 10$. Is the collected data enough to reject the null hypothesis $H_0: \mu_{\text{phone}} > 1120$, which claims that the new phone has the longest average battery lifetime? To answer this, calculate your estimate $\hat{\mu}_{\text{phone}}$ and perform the corresponding test with 7% significance. Provide the test statistic or the p-value to support your conclusion.
- 2) Still assuming that $\sigma = 10$: what is the minimum level of significance to reject the company's claim?
- 3) Assume now that the website doesn't know the value of σ . Calculate an estimate for σ and perform the corresponding test with 7% significance. Can you reject the claim of the company in this case? Provide the test statistic or the p-value to support your conclusion.